SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade.

Recognised as Scientific and Industrial Research Organisation

R MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., IN

Regulation: R23	**IV / IV - B.Tech. I - Semester**

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

COURSE STRUCTURE

(With effect from 2023-24 admitted Batch onwards)

Course Code	Course Name	Category	L	T	P	Cr	C.I.E	S.E.E	Total Marks
B23AM4101	Generative AI	**PC**	3	0	0	3	30	70	100
B23HS4101	Human Resource & Project Management	**HS**	2	0	0	2	30	70	100
#PE-IV	Professional Elective-IV	**PE**	3	0	0	3	30	70	100
#PE-V	Professional Elective-V	**PE**	3	0	0	3	30	70	100
#OE-III	Open Elective-III	**OE**	3	0	0	3	30	70	100
#OE-IV	Open Elective-IV	**OE**	3	0	0	3	30	70	100
B23AM4111	Prompt Engineering / SWAYAM Plus - Certificate program in Prompt Engineering and Chat GPT	**SEC**	0	1	2	2	30	70	100
B23MC4101	Constitution of India	**MC**	2	0	0	-	30	-	30
B23AM4112	Evaluation of Industry Internship / Mini Project	**PR**	-	-	-	2		50	50
		Total	**19**	**1**	**02**	**21**	**240**	**540**	**780**

	Course Code	Course
#PE-IV	B23AM4102	Quantum Computing
	B23AM4103	Cloud Computing
	B23CS4103	Block chain Technology
	B23AM4104	Software Project Management
	B23AM4105	12 week MOOC Swayam /NPTEL course recommended by the BoS
#PE-V	B23CS4107	Agile Methodologies
	B23AM4106	High Performance Computing
	B23AM4107	Explainable AI
	B23AM4108	Agentic AI
	B23AM4109	Edge AI
	B23AM4110	12 week MOOC Swayam/NPTEL course recommended by the BoS
#OE-III	Student has to study one open elective offered by CE or MECH or ECE or EEE or S&H from the list enclosed.	
#OE-IV	Student has to study one open elective offered by CE or MECH or ECE or EEE or S&H from the list enclosed.	

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AM4101	**PC**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

GENERATIVE AI

(For AI & ML)

Course Objectives:

The objectives of this course are

1.	To introduce the fundamental concepts, evolution, and significance of Generative AI and its applications.
2.	To familiarize students with generative models for text, image, and multimodal content generation
3.	To enable students to apply generative AI techniques for content generation using modern architectures such as transformers, GANs, and diffusion models.
4.	To provide exposure to prompt engineering, fine-tuning, and open-source frameworks for building Generative AI applications.
5.	To create awareness about ethical, responsible, and real-world implications of Generative AI systems.

Course Outcomes: At the end of the course, Students will be able to

S. No.	Outcome	Knowledge Level
1.	Use Generative AI concepts and models to suitable applications and use cases	**K3**
2.	Apply transformers, LLMs, prompting, and retrieval techniques for language generation	**K3**
3.	Apply GANs, VAEs, diffusion, and transformer-based models for image generation	**K3**
4.	Analyze GAN-based techniques for painting and music generation	**K4**
5.	Apply open-source models and frameworks for fine-tuning and deployment of Generative AI applications	**K3**

SYLLABUS

UNIT-I (10Hrs)	**Introduction to Gen AI**: Historical Overview of Generative modelling, Difference between Gen AI and Discriminative Modeling, Importance of generative models in AI and Machine Learning, Types of Generative models, GANs, VAEs, autoregressive models and Vector quantized Diffusion models, Understanding if probabilistic modeling and generative process, Challenges of Generative Modeling, Future of Gen AI, Ethical Aspects of AI, Responsible AI, Use Cases.
UNIT-II	**Generative Models for Text**: Language Models Basics, Building blocks of Language

(10 Hrs)	models, Transformer Architecture, Encoder and Decoder, Attention mechanisms, Generation of Text, Models like BERT and GPT models, Exploring ChatGPT, Prompt Engineering: Designing Prompts, Revising Prompts using Reinforcement Learning from Human Feedback (RLHF), Retrieval Augmented Generation, Multimodal LLM, Issues of LLM like hallucination.
UNIT-III (10 Hrs)	**Generation of Images**: Introduction to Generative Adversarial Networks, Adversarial Training Process, Nash Equilibrium, Variational Auto encoders, **Encoder-Decoder Architectures**: Stable Diffusion Models, Introduction to Transformer-based Image Generation, CLIP, Visual Transformers ViT- Dall-E2 and Dall-E3, GPT-4V, Issues of Image Generation models like Mode Collapse and Stability
UNIT-IV (10 Hrs)	**Generation of Painting, Music, and Play**: Variants of GAN, Types of GAN, Cyclic GAN, Using Cyclic GAN to Generate Paintings, Neural Style Transfer, Style Transfer, Music Generating RNN, MuseGAN
UNIT-V (10 Hrs)	**Open Source Models and Programming Frameworks**: Training and Fine tuning of Generative models, GPT4All, Transfer learning and Pretrained models, Google Copilot, Programming LLM, LangChain, Open Source Models, Llama, Programming for Transformer, Deployment, Hugging Face AutoTrain- Training vision models without coding

Textbooks:

1.	Denis Rothman, "Transformers for Natural Language Processing and Computer Vision", Third Edition, Packt Books, 2024
2.	Altaf Rehmani, "Generative AI for Everyone", First Edition, BlueRose One, 2024.

Reference Books:

1.	David Foster," Generative Deep Learning", Second Edition, O'Reily Books, 2024.
2.	Karthikeyan Sabesan, Sivagamisundari, and Nilip Dutta, *Generative AI for Everyone: Deep Learning, NLP, and LLMs for Creative and Practical Applications*, First Edition, BPB Publications,2025

e-Resources

1.	microsoft/generative-ai-for-beginners, https://github.com/microsoft/generative-ai-for-beginners?utm_source=chatgpt.com
2.	Generative AI Research Resources, https://midas.umich.edu/research/generative-ai-hub/generative-ai-research-resources/
3	https://www.coursera.org/learn/generative-ai-for-everyone#modules
4	https://poloclub.github.io/ganlab/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS4101	**HS**	**2**	**--**	**--**	**2**	**30**	**70**	**3 Hrs.**

HUMAN RESOURCE & PROJECT MANAGEMENT

(Common to CSE, AIML, IT, AIDS, CSG, CSIT)

Course Objectives:

1.	Explain the nature, scope, functions of HRM, and processes of HR planning, recruitment, and selection.
2.	Explain HRD concepts, training methods, performance appraisal systems, and career development practices.
3.	Explain the basic concepts of project management, including project life cycle, resource management, and monitoring techniques.
4.	Explain project selection and appraisal methods using financial and strategic evaluation tools.
5.	Explain project implementation processes, organizational structures, and performance evaluation methods.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	Explain HRM functions, HR planning, and recruitment–selection procedures.	K2
2.	Explain HRD practices, training techniques, and performance appraisal methods.	K2
3.	Explain project management concepts, project networks, and monitoring processes.	K2
4.	Explain project appraisal techniques such as SWOT analysis, payback period, and NPV.	K2
5.	Explain project implementation strategies, organizational forms, and performance evaluation processes.	K2

SYLLABUS

UNIT-I (10Hrs)	**HRM:** Nature, Scope, Concept of HRM - Functions of HRM - Role of HR manager – Emerging trends in HRM – E-HRM - HR audit models – ethical aspects of HRM. HR Planning – Job Design- Recruitment - Sources of recruitment -Selection- Selection Procedure.
UNIT-II (10 Hrs)	**HRD**: HR accounting – Models - Concept of Training and Development –- Methods of Training. Performance Appraisal: Importance – Methods of performance appraisal – Career Development and Counseling – group interaction.
UNIT-III (10 Hrs)	**Basics of Project Management:** Concept–resource management- Project environment – Types of Projects –project networks-DPR- Project life cycle – Project proposals –

	Monitoring project progress – Project appraisal and Project selection-80-20 rules.
UNIT-IV (10 Hrs)	**Project Selection and Appraisal:** Brainstorming and concept evolution, Project selection and evaluation, Selection criteria and models, Types of appraisals, SWOT analysis, Cash flow analysis, Payback period, and Net present value.
UNIT-V (10 Hrs)	**Project Implementation and Review:** Identify various project types and apply appropriate management strategies for each. Forms of project organization – project planning – project control – human aspects of project management – prerequisites for successful project implementation – project review.
Textbooks:	
1.	Sharon Pande and Swapnalekha Basak, Human Resource Management, Text and Cases, Vikas Publishing, 2e, 2016.
2.	K. Nagarajan, Project Management, New Age International Publishers, 8e, 2017
Reference Books:	
1.	Subba Rao P: ―Personnel and Human Resource Management-Text and Cases‖, Himalaya Publications, Mumbai, 2013.
2.	Prasanna Chandra, ―Projects, Planning, Analysis, Selection, Financing Implementation and Review‖, Tata McGraw Hill Company Pvt. Ltd., New Delhi 1998
3.	Vasanth Desai, Project Management, 4th edition, Himalaya Publications 2018.
4.	Lalithabalakrishnan and Gowri, Project Management, Himalaya publishing house, New Delhi 2022
5.	K Aswathappa: ―Human Resource and Personnel Management‖, Tata McGraw Hill, New Delhi, 2013
e-Resources	
1.	Robert L. Mathis, John H. Jackson, Manas Ranjan Tripathy, Human Resource Management, Cengage Learning 2016.
2.	Stewart R. Clegg, Torgeir Skyttermoen, Anne Live Vaagaasar, Project Management, Sage Publications, 1e, 2021.

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23AM4102	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

QUANTUM COMPUTING
(For AIML)

Pre-requisites: Linear Algebra, Digital Logic Design.

Course Objectives: Students are expected

1.	Articulate the significance of quantum computing in solving problems intractable for classical systems.
2.	Apply fundamental concepts of linear algebra, Hilbert spaces, and probability theory to quantum systems.
3.	Construct quantum circuits and generate Bell states to demonstrate entanglement.
4.	Evaluate the advantages and limitations of quantum algorithms in solving real-world problems.
5.	Compare classical and quantum information theory, including fault-tolerant computation.

Course Out Comes: At the end of the course students will be able to

S. No	OUTCOME	KL
1.	Apply quantum computing concepts to differentiate bits vs qubits and classical vs quantum operations.	K3
2.	Apply principles of linear algebra and quantum mechanics to model quantum states and measurements in Hilbert space.	K3
3.	Apply quantum gates and circuits to manipulate qubits and create Bell states.	K3
4.	Analyze the efficiency and computational complexity of quantum algorithms compared to classical counterparts.	K4
5.	Apply quantum error correction and cryptography principles to ensure secure information transmission.	K3

SYLLABUS

UNIT-I (10 Hrs)	**History and Introduction to Quantum Computing:** History of Quantum Computing: Importance of Mathematics, Physics and Biology, Introduction to Quantum Computing: Bits Vs Qubits, Classical Vs Quantum logical operations
UNIT-II (10 Hrs)	**Background Concepts:** Background Mathematics: Basics of Linear Algebra, Hilbert space, Probabilities and measurements, **Background Physics:** Pauli's exclusion Principle, Superposition, Entanglement and supersymmetry, Density operators and correlation, Basics of quantum mechanics, Measurements in bases other than computational basis, **Background Biology:** Basic concepts of Genomics and Proteomics (Central Dogma)
UNIT-III (10 Hrs)	**Qubit and Quantum Circuits**: Qubit: Physical implementations of Qubit, Qubit as a quantum unit of information, The Bloch sphere, Quantum Circuits: Single qubit gates,

	Multiple qubit gates, Designing the quantum circuits, Bell states
UNIT-IV (10 Hrs)	**Quantum Algorithms:** Classical computation on quantum computers, Relationship between quantum and classical complexity classes, Deutsch's algorithm, Deutsch's-Jozsa algorithm, Shor's factorization algorithm, Grover's search algorithm
UNIT-V (10 Hrs)	**Noise, Error Correction, and Quantum Information:** Noise and error correction: Graph states and codes, Quantum error correction, Fault-tolerant computation, **Quantum Information and Cryptography:** Comparison between classical and quantum information theory, Quantum Cryptography, Quantum teleportation
TEXTBOOK:	
1.	Nielsen M. A., Quantum Computation and Quantum Information, Cambridge, 10th Anniversary Edition, 2010
REFERENCE BOOKS:	
1.	Quantum Computing for Computer Scientists by Noson S. Yanofsky and Mirco A. Mannucci, 2008
2.	Benenti G., Casati G. and Strini G., Principles of Quantum Computation and Information, Vol. I, Basic Concepts, Vol II: Basic Tools and Special Topics, World Scientific, 2007.
3.	Pittenger A. O., An Introduction to Quantum Computing Algorithms, Birkhauser Verlag AG, V19, 2000.
E-Resources:	
1.	History and Fundamentals of Quantum Computing: https://plato.stanford.edu/entries/qt-quantcomp/ (Stanford Encyclopedia of Philosophy)
2.	Mathematical, Physical, and Biological Foundations for Quantum Computing: https://quantum-computing.ibm.com/composer/docs/iqx/guide/quantum-background (IBM Quantum)
3.	Qubits and Quantum Circuit Design: https://qiskit.org/learn/course/basics-qubit/ (Qiskit (IBM))
4.	Core Quantum Algorithms and Complexity: https://www.nature.com/articles/nature23459 (Nature)
5.	Quantum Error Correction, Cryptography, and Teleportation: https://errorcorrectionzoo.org/ (Error Correction Zoo (Caltech))

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AM4103	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

CLOUD COMPUTING

(AI&ML)

Course Objectives:

1.	**Understanding** cloud computing models, architectures, and core services along with their deployment strategies.
2.	**Exposure** to enabling technologies such as distributed computing, virtualization, and containers in cloud environments.
3.	**Familiarity** with technical challenges and security considerations in cloud adoption.
4.	**Awareness** of advanced trends in cloud computing including serverless, edge, and fog computing.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	Apply cloud architecture, service models, and deployment models to practical computing scenarios.	**K3**
2.	Analyze the role of distributed and parallel architectures in cloud computing.	**K4**
3.	Demonstrate the use of virtualization and containers in cloud resource provisioning.	**K3**
4.	Analyze technical challenges in cloud computing such as security, scalability, and interoperability.	**K4**
5.	Use serverless and edge/fog computing concepts to optimize performance and cost-efficiency in cloud-based systems.	**K3**

SYLLABUS

UNIT-I (10Hrs)	**Introduction to Cloud Computing Fundamentals** Cloud computing at a glance, defining a cloud, cloud computing reference model, types of services (IaaS, PaaS, SaaS), cloud deployment models (public, private, hybrid), utility computing, cloud computing characteristics and benefits, cloud service providers (Amazon Web Services, Microsoft Azure, Google AppEngine).
UNIT-II (10 Hrs)	**Cloud Enabling Technologies** Ubiquitous Internet, parallel and distributed computing, elements of parallel computing, hardware architectures for parallel computing (SISD, SIMD, MISD, MIMD), elements of distributed computing, Inter-process communication, technologies for distributed computing, remote procedure calls (RPC), service-oriented architecture (SOA), web services, virtualization.

UNIT-III (10 Hrs)	**Virtualization and Containers** Characteristics of virtualized environments, taxonomy of virtualization techniques, virtualization and cloud Computing, pros and cons of virtualization, technology examples (XEN, VMware), building blocks of containers, container platforms (LXC, Docker), container orchestration, Docker Swarm and Kubernetes, public cloud VM (e.g. Amazon EC2) and container (e.g. Amazon Elastic Container Service) offerings.
UNIT-IV (10 Hrs)	**Cloud computing challenges** Economics of the cloud, cloud interoperability and standards, scalability and fault tolerance, energy efficiency in clouds, federated clouds, cloud computing security, fundamentals of computer security, cloud security architecture, security in cloud deployment models.
UNIT-V (10 Hrs)	**Advanced concepts in cloud computing** Serverless computing, Function-as-a-Service, serverless computing architecture, public cloud (e.g. AWS Lambda) and open-source (e.g. OpenFaaS) serverless platforms, edge and fog computing.

Textbooks:

1.	Mastering Cloud Computing, 2nd edition, Rajkumar Buyya, Christian Vecchiola, Thamarai Selvi, Shivananda Poojara, Satish N. Srirama, Mc Graw Hill, 2024.
2.	Distributed and Cloud Computing, Kai Hwang, Geoffery C. Fox, Jack J. Dongarra, Elsevier, 2012.

Reference Books:

1.	Cloud Computing, Theory and Practice, Dan C Marinescu, 2nd edition, MK Elsevier, 2018.
2.	Essentials of cloud Computing, K. Chandrasekhran, CRC press, 2014.
3.	Online documentation and tutorials from cloud service providers (e.g., AWS, Azure, GCP)
4.	Docker, Reference documentation, https://docs.docker.com/reference/
5.	OpenFaaS, Serverless Functions Made Simple, https://docs.openfaas.com/

e-Resources

1.	https://onlinecourses.nptel.ac.in/noc21_cs14/preview
2.	https://onlinecourses.nptel.ac.in/noc24_cs17/preview

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CS4103	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

BLOCKCHAIN TECHNOLOGY
(Common to CSE, AIML)

Pre-requisites: Strong foundation in DS,CNS,CN

Course Objectives:

1.	To learn the fundamentals of Block Chain and various types of block chain and consensus mechanism.
2.	To understand public block chain system, Private block chain system and consortium block chain.
3.	Able to know the security issues of blockchain technology.

Course Outcomes

S.No	Outcome	KL
1.	Explain the fundamental concepts of the Block chain technology.	K2
2.	Demonstrate public block chain system concepts and the smart contracts.	K2
3.	Demonstrate private and consortium block chain systems and Initial Coin Offering.	K2
4.	Apply block chain concepts and security to the different applications.	K3
5.	Develop block chain programs and chain codes using python and Hyperledger fabric	K3

SYLLABUS

UNIT-I (10Hrs)	**Fundamentals of Blockchain:** Introduction, Origin of Blockchain, Blockchain Solution, Components of Blockchain, Block in a Blockchain, The Technology and the Future. **Blockchain Types and Consensus Mechanism:** Introduction, Decentralization and Distribution, Types of Blockchain, Consensus Protocol. **Cryptocurrency:** Bitcoin, Altcoin and Token: Introduction, Bitcoin and the Cryptocurrency, Cryptocurrency Basics, Types of Cryptocurrencies, Cryptocurrency Usage.
UNIT-II (10 Hrs)	**Public Blockchain System:** Introduction, Public Blockchain, Popular Public Blockchains, The Bitcoin Blockchain, Ethereum Blockchain. **Smart Contracts:** Introduction, Smart Contract, Characteristics of a Smart Contract, Types of Smart Contracts, Types of Oracles, Smart Contracts in Ethereum, Smart Contracts in Industry.
UNIT-III (10 Hrs)	**Private Blockchain System:** Introduction, Key Characteristics of Private Blockchain, Private Blockchain, Private Blockchain Examples, Private Blockchain and Open Source, E- commerce Site Example, Various Commands (Instructions) in E-commerce

	Blockchain, Smart Contract in Private Environment, State Machine, Different Algorithms of Permissioned Blockchain, Byzantine Fault, Multichain. **Consortium Blockchain:** Introduction, Key Characteristics of Consortium Blockchain, Need of Consortium Blockchain, Hyperledger Platform, Overview of Ripple, Overview of Corda. **Initial Coin Offering:** Introduction, Blockchain Fundraising Methods, Launching an ICO, Investing in an ICO, Pros and Cons of Initial Coin Offering, Successful Initial Coin Offerings, Evolution of ICO, ICO Platforms.
UNIT-IV (10 Hrs)	**Security in Blockchain**: Introduction, Security Aspects in Bitcoin, Security and Privacy Challenges of Blockchain in General, Performance and Scalability, Identity Management and Authentication, Regulatory Compliance and Assurance, Safeguarding Blockchain Smart Contract (DApp), Security Aspects in Hyperledger Fabric. **Applications of Blockchain:** Introduction, Blockchain in Banking and Finance, Blockchain in Education, Blockchain in Energy, Blockchain in Healthcare, Blockchain in Real-estate, Blockchain in Supply Chain, The Blockchain and IoT. Limitations and Challenges of Blockchain.
UNIT-V (10 Hrs)	**Blockchain Platform using Python:** Introduction, Learn How to Use Python Online Editor, Basic Programming Using Python, Python Packages for Blockchain. **Blockchain platform using Hyperledger Fabric:** Introduction, Components of Hyperledger Fabric Network, Chain codes from Developer.ibm.com, Blockchain Application Using Fabric Java SDK. **Blockchain Case Studies:** Retail, Banking and Financial Services, Healthcare, Energy and Utilities.

Textbooks:	
1.	"Block chain Technology", Chandramouli Subramanian, Asha A.George, Abhilasj K A and Meena Karthikeyan, Universities Press, First Edition, 2020.
Reference Books:	
1.	Blockchain Blue print for Economy, Melanie Swan, SPD Oreilly.
2.	Blockchain for Business, Jai Singh Arun, Jerry Cuomo, Nitin Gauar, Pearson Addition Wesley.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AM4104	**PE**	**3**	**--**	**3**	**3**	**30**	**70**	**3 Hrs.**

SOFTWARE PROJECT MANAGEMENT

(For AIML)

Course Objectives:

1.	To describe and determine the purpose and importance of project management from the perspectives of planning, tracking and completion of project
2.	To compare and differentiate organization structures and project structures
3.	To implement a project to manage project schedule, expenses and resources with the application of suitable project management tools

Course Outcomes: At the end of the course, Students will be able to

S.No	Outcome	KnowledgeLevel
1.	Apply the process to be followed in the software development life-cycle models	**K3**
2.	Apply the concepts of project management & planning	**K3**
3.	Analyze the project plans through managing people, communications and change	**K4**
4.	Conduct activities necessary to successfully complete and close the Software projects	**K4**
5.	Analyze communication, modeling, and construction & deployment practices in software development	**K4**

SYLLABUS

UNIT-I (10Hrs)	**Conventional Software Management**: The waterfall model, conventional software Management performance. **Evolution of Software Economics**: Software Economics, pragmatic software cost estimation. **Improving Software Economics**: Reducing Software product size, improving software processes, improving team effectiveness, improving automation, Achieving required quality, peer inspections. **The old way and the new**: The principles of conventional software Engineering, principles of modern software management, transitioning to an iterative process.
UNIT-II (10 Hrs)	**Life cycle phases**: Engineering and production stages, inception, Elaboration, construction, transition phases. **Artifacts of the process**: The artifact sets, Management artifacts, Engineering artifacts, programmatic artifacts.
UNIT-III (10 Hrs)	**Model based software architectures**: A Management perspective and technical perspective. **Work Flows of the process**: Software process workflows, Iteration

	workflows. **Checkpoints of the process**: Major mile stones, Minor Milestones, Periodic status assessments. **Iterative Process Planning**: Work breakdown structures, planning guidelines, cost and schedule estimating, Iteration planning process, Pragmatic planning.
UNIT-IV (10 Hrs)	**Project Organizations and Responsibilities**: Line-of-Business Organizations, Project Organizations, evolution of Organizations. **Process Automation**: Automation Building blocks, The Project Environment. **Project Control and Process instrumentation**: The seven core Metrics, Management indicators, quality indicators, life cycle expectations, pragmatic Software Metrics, Metrics automation.
UNIT-V (10 Hrs)	Agile Methodology, Adapting to Scrum, Patterns for Adopting Scrum, Iterating towards Agility. **Fundamentals of DevOps**: Architecture, Deployments, Orchestration, Need, Instance of applications, DevOps delivery pipeline, DevOps eco system. DevOps adoption in projects: Technology aspects, Agiling capabilities, Tool stack implementation, People aspect, processes

Textbooks:

1.	Software Project Management, Walker Royce, PEA, 2005
2.	Succeeding with Agile: Software Development Using Scrum, Mike Cohn, Addison Wesley,2009.
3.	The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations, Gene Kim , John Willis , Patrick Debois , Jez Humb,1st Edition, O'Reilly publications, 2016.

Reference Books:

1.	Software Project Management, Bob Hughes,3/e, Mike Cotterell, TMH,2002.
2.	Software Project Management, Joel Henry, PEA,2004.
3.	Software Project Management in practice, Pankaj Jalote, PEA, 2005
4.	Effective Software Project Management, Robert K.Wysocki, Wiley,2006
5.	Project Management in IT, Kathy Schwalbe, Cengage,2000.

e-Resources

1.	Software Project Management, IIT Kharagpur Prof. Rajib Mall, Prof. Durga Prasad Mohapatra https://nptel.ac.in/courses/106105218
2.	Affective Computing, IIIT Delhi Prof. Jainendra Shukla Prof. Abhinav Dhall https://nptel.ac.in/courses/106106244

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CS4107	PE	3	--	--	3	30	70	3 Hrs.

AGILE METHODOLOGIES

(Common to CSE, AIML, AIDS, CSBS, IT)

Pre-requisites: Software Engineering

Course Objectives: Students are expected

1.	Gain deep comprehension of Agile philosophy, principles, and values to support high-quality software delivery.
2.	Employ different Agile frameworks (Scrum, XP, Lean, Kanban) to real-world projects for improved adaptability, productivity, and customer satisfaction.

Course Out Comes: At the end of the course students will be able to

S. No	OUTCOME	Knowledge Level
1.	Apply agile values and principles to foster team collaboration and adaptability in projects.	K3
2.	Demonstrate the 12 agile principles to direct project delivery in changing contexts.	K3
3.	Use Scrum roles, events, and artifacts to oversee iterative development cycles.	K3
4.	Model XP practices to boost software quality and team responsiveness.	K3
5.	Determine Lean and Kanban principles to detect waste and streamline development workflows.	K3

SYLLABUS

UNIT-I (10 Hrs)	**Learning Agile:** Agile, Getting Agile into your brain, **Understanding Agile values**: No Silver Bullet, Agile to the Rescue. A fractured perspective, The Agile Manifesto, Understanding the Elephant, Where to Start with a New Methodology.
UNIT-II (10 Hrs)	**The Agile Principles:** The 12 Principles of Agile Software, The Customer Is Always Right, Delivering the Project, Better Project Delivery for the Ebook Reader Project. Communicating and Working Together, Project Execution—Moving the Project Along, Constantly Improving the Project and the Team. The Agile Project: Bringing All the Principles Together
UNIT-III (10 Hrs)	**SCRUM and Self-Organizing Teams:** The Rules of Scrum, Act I: I Can Haz Scrum, Everyone on a Scrum Team owns the Project, Status Updates Are for Social Networks!, The Whole Team Uses the Daily Scrum, Feedback and the Visibility-Inspection-

	Adaptation Cycle, The Last Responsible Moment, Sprinting into a Wall, Sprints, Planning, and Retrospectives. **Scrum Planning And Collective Commitment:** Not Quite Expecting the Unexpected, User Stories, Velocity, and Generally Accepted Scrum Practices, Victory Lap, Scrum Values Revisited.
UNIT-IV (10 Hrs)	**XP And Embracing Change:** Going into Overtime, The Primary Practices of XP, The Game Plan Changed, but We're Still Losing, The XP Values Help the Team Change Their Mindset, An Effective Mindset Starts with the XP Values, The Momentum Shifts, Understanding the XP Principles Helps You Embrace Change. **XP, Simplicity, and Incremental Design:** Code and Design, Make Code and Design Decisions at the Last Responsible Moment, Final Score.
UNIT-V (10 Hrs)	**Lean, Eliminating Waste, and Seeing the whole:** Lean Thinking, Creating Heroes and Magical Thinking. Eliminate Waste, Gain a Deeper Understanding of the Product, Deliver As Fast As Possible. **Kanban, Flow, and Constantly Improving:** The Principles of Kanban, Improving Your Process with Kanban, Measure and Manage Flow, Emergent Behavior with Kanban. **The Agile Coach:** Coaches Understand Why People Don't Always Want to Change. The Principles of Coaching.
Textbook:	
1.	Andrew Stellman, Jennifer Greene, Learning Agile, O'Reilly, 2015.
Reference books:	
1.	Andrew Stellman, Jennifer Green, Head first Agile, O'Reilly, 2017.
2.	Rubin K , Essential Scrum : A Practical Guide To The Most Popular Agile Process, Addison-Wesley, 2013
E-Resources:	
1.	https://www.agilealliance.org
2.	https://www.scrum.org
3.	https://www.scrumtrainingseries.com
4.	https://www.agilemanifesto.org
5.	https://www.scrumguides.org

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AM4106	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

HIGH PERFORMANCE COMPUTING

(For AIML)

Course Objectives:

1.	To understand the architectural transitions from single-core to multi-core and GPU systems, focusing on the physical and logical limitations of memory and power.
2.	To master the techniques of decomposing complex problems (specifically in sorting, graphs, and matrices) into tasks that can execute concurrently.
3.	To gain hands-on proficiency in shared-memory programming (OpenMP) and massively parallel GPU programming (CUDA) for AI-driven applications.

Course Outcomes: By the end of this course, students will be able to:

S.No	Outcome	Knowledge Level
1.	Analyze the evolution of microprocessor architectures and identify communication bottlenecks and scalability constraints in parallel hardware platforms.	**K4**
2.	Apply decomposition and mapping techniques to design efficient parallel algorithm models while minimizing interaction overheads.	**K3**
3.	Construct multi-threaded software solutions by integrating OpenMP and collective communication patterns like Scatter, Gather, and All-Reduce.	**K3**
4.	Demonstrate the performance of parallel systems using analytical models (Speedup, Efficiency, Isoefficiency) and optimize dense matrix kernels.	**K3**
5.	Examine the impact of CUDA memory management and synchronization strategies on the scalability of high-performance parallel solutions.	**K4**

SYLLABUS

UNIT-I (10Hrs)	**INTRODUCTION:** Motivating Parallelism, Scope of Parallel Computing, Parallel Programming Platforms: Implicit Parallelism, Trends in Microprocessor and Architectures, Limitations of Memory, System Performance, Dichotomy of Parallel Computing Platforms, Physical Organization of Parallel Platforms, Communication Costs in Parallel Machines, Scalable design principles, Architectures: N-wide superscalar architectures, Multi-core architecture.
UNIT-II (10 Hrs)	**PARALLEL PROGRAMMING:** Principles of Parallel Algorithm Design: Preliminaries, Decomposition Techniques, Characteristics of Tasks and Interactions, Mapping

	Techniques for Load Balancing, Methods for Containing Interaction Overheads, Parallel Algorithm Models, The Age of Parallel Processing, the Rise of GPU Computing, A Brief History of GPUs, Early GPU.
UNIT-III (10 Hrs)	**BASIC COMMUNICATION:** Operations- One-to-All Broadcast and All-to-One Reduction, All-to-All Broadcast and Reduction, All-Reduce and Prefix-Sum Operations, Scatter and Gather, All-to-All Personalized Communication, Circular Shift, Improving the Speed of Some Communication Operations. Programming shared address space platforms: threads- basics, synchronization, OpenMP programming
UNIT-IV (10 Hrs)	**ANALYTICAL MODELS**: Sources of overhead in Parallel Programs, Performance Metrics for Parallel Systems, and The effect of Granularity on Performance, Scalability of Parallel Systems, Minimum execution time and minimum cost, optimal execution time. **Dense Matrix Algorithms**: MatrixVector Multiplication, Matrix-Matrix Multiplication
UNIT-V (10 Hrs)	**PARALLEL ALGORITHMS- SORTING AND GRAPH :** Issues in Sorting on Parallel Computers, Bubble Sort and its Variants, Parallelizing Quick sort, All-Pairs Shortest Paths, Algorithm for sparse graph, Parallel Depth-First Search, Parallel BestFirst Search. **CUDA Architecture**: CUDA Architecture, Using the CUDA Architecture, Applications of CUDA Introduction to CUDA C-Write and launch CUDA C kernels, Manage GPU memory, Manage communication and synchronization, Parallel programming in CUDA-C.

Textbooks:	
1.	Ananth Grama, Anshul Gupta, George Karypis, and Vipin Kumar, "Introduction to Parallel Computing", 2nd edition, Addison-Wesley, 2003, ISBN: 0-201-64865-2. [Unit 1 to 4]
2.	Jason sanders, Edward Kandrot, “CUDA by Example”, Addison-Wesley, 1st Edition-2010. ISBN-13: 978-0-13-138768-3. [Unit-5]
Reference Books:	
1.	Kai Hwang, ”Scalable Parallel Computing”, McGraw Hill 1998, ISBN:0070317984
2.	Shane Cook, “CUDA Programming: A Developer's Guide to Parallel Computing with GPUs”, Morgan Kaufmann (Elsevier) Publishers Inc. San Francisco, CA, USA,1st Edition-2013, ElsevierISBN: 9780124159884.
3.	David Culler Jaswinder Pal Singh, ”Parallel Computer Architecture: A Hardware/Software Approach”, Morgan Kaufmann (Elsevier),1st Edition-1999, ISBN 978-1-55860-343-1
4.	Rod Stephens, “Essential Algorithms”, Wiley, 1st Edition-2013, ISBN: 978-1-118-61210-1.
5.	Derek Lloyd, “High Performance GPU Computing with C++ and CUDA: Design Robust, Hardware-Agnostic Solutions for the Next Generation of Accelerated Computing”,1st Edition-2025. ISBN-13: 979-8275358292.

e-Resources	
1.	https://onlinecourses.nptel.ac.in/noc26_cs32/preview
2.	https://www.youtube.com/watch?v=RpT-fRbQeuM

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AM4107	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

EXPLAINABLE AI
(For AIML)

Course Objectives:

1.	Understand the importance of explainability in AI and its impact on stakeholders.
2.	Explore different techniques and methods for making AI Systems Explainable.
3.	Explain the trade-offs between model complexity and interpretability.
4.	Examine the ethical and societal implications of XAI.
5	Apply XAI techniques to real-world datasets and scenarios.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	**Understand** the fundamental concepts of Explainable AI (XAI) and its role in interpreting machine learning model predictions.	K2
2.	**Use** interpretable modeling techniques such as linear models, decision trees, rule-based systems, and neural network-based approaches to explain and analyze model predictions.	K3
3.	**Apply** various Explainable AI methods such as PDP, ICE, LIME, SHAP, saliency maps, and integrated gradients to interpret and analyze linear, non-linear, and deep learning models.	K3
4.	**Apply** evaluation techniques and metrics used to assess the effectiveness, trustworthiness, and ethical considerations of XAI methods.	K3
5.	**Apply** Explainable AI techniques in diverse application domains such as healthcare, finance, autonomous systems, time series forecasting, natural language processing, and computer vision to interpret model outcomes.	K3

SYLLABUS

UNIT-I (10 Hrs)	**Introduction to Explainable AI (XAI):** Motivations for XAI, Importance of Interpretability and Transparency techniques for XAI, Model-specific interpretability methods (e.g., decision trees, rule-based systems), Model-agnostic interpretability methods (e.g., LIME, SHAP), Post-hoc explanation techniques (e.g., feature importance, counterfactual explanation).
UNIT-II (10 Hrs)	**Interpretable Models:** Linear Models, Decision Trees and Rule-based systems, Symbolic AI approaches,

	Interpretable Neural Networks, Sparse Neural Networks, Attention mechanisms, Layer-wise relevance propagation (LRP).
UNIT-III (10 Hrs)	**Methods for Explainable AI:** Partial Dependence Plot (PDP), Conformal Prediction, Individual Conditional Expectation (ICE), Feature Importance, Saliency Maps, Local Interpretable Model Agnostic Explanations (LIME), SHAP, Integrated Gradient (IG), Explainability for Linear Models, Non-Linear Models and Deep Learning Models.
UNIT-IV (10 Hrs)	**Evaluation of XAI Methods:** Quantitative metrics for interpretability, Human-centric evaluation methods, Ethical and Societal Implications of XAIB, Trust and accountability in AI systems, Regulatory considerations.
UNIT-V (10 Hrs)	**Applications of XAI:** Healthcare (e.g., Medical diagnosis, personalized treatment), Finance (e.g., credit scoring, fraud detection), Autonomous systems (e.g., self-driving cars, drones), Explainability in Time Series forecasting, Natural Language Processing and Computer Vision.
Textbooks:	
1.	“Interpretable machine learning. A Guide for Making Black Box Models Explainable” by Molnar, Christop, 2019. https://christophm.github.io/interpretable-ml-book/.
2.	Munn, Michael, and David Pitman. Explainable AI for practitioners. “O'Reilly Media, Inc”.2022.
3.	Explainable AI: Interpreting, Explaining and Visualizing Deep Learning, by Wojciech Samek, Grégoire Montavon, Andrea Vedaldi, Lars Kai Hansen, and Klaus-Robert Müller, 1st Edition, 2019, Springer Nature, Lecture Notes in Computer Science (LNCS), Volume 11700, ISBN: 978-3-030-28953-9.
Reference Books:	
1.	https://doi.org/10.1016/j.iswa.2026.200647
e-Resources	
1.	https://github.com/christophM/interpretable-ml-book
2.	https://ai.google/responsibilities/responsible-ai/
3.	https://arxiv.org/html/2410.17139v2#S3

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AM4108	PE	3	--	--	3	30	70	3 Hrs.

AGENTIC AI

(For AIML)

Course Objectives:

1.	To provide a strong foundation in the fundamental concepts, architectures, and evolution of Agentic AI systems, and to distinguish them from traditional, reactive, and generative AI models.
2.	To enable students to apply decision-making, planning, reinforcement learning, and Large Language Model (LLM)-based techniques for designing intelligent, goal-oriented autonomous agents.
3.	To develop the ability to design, implement, and evaluate single-agent and multi-agent systems using modern frameworks, while addressing ethical, safety, and societal considerations.

Course Outcomes: At the end of the course, Students will be able to

S.No	Outcomes	Knowledge Level
1.	Explain the fundamental concepts, characteristics, and architectures of Agentic AI systems, and differentiate them from traditional, reactive, and generative AI models	**K2**
2.	Apply decision-making, planning, and reinforcement learning techniques to design autonomous agents capable of goal-directed behavior under deterministic and uncertain environments.	**K3**
3.	Analyze the role of Large Language Models (LLMs) in agentic systems and implement prompt- driven reasoning, memory, and tool-use mechanisms to enable intelligent agent behavior.	**K4**
4.	Apply agentic AI frameworks to design and implement multi-agent systems with memory, communication, and scalable deployment strategies.	**K3**
5.	Evaluate ethical, safe, and responsible design of agentic AI systems considering bias , privacy, and real-world applications.	**K4**

SYLLABUS

UNIT-I (10Hrs)	**Foundations of Agentic Intelligence**: Understand what Agentic AI is, how it differs from traditional and generative AI, and learn its core theoretical building blocks. Wat is Agentic AI vs. **Generative/Reactive AI**; Autonomy, Proactivity, and Goal-Directed Behavior; Components of an intelligent agent: Perception, Reasoning, Learning, Action; Agent architectures: Reactive, Deliberative, Hybrid; Rationality and decision models; Real-world use cases and motivation for agentic systems.

UNIT-II (10 Hrs)	**Decision Making, Planning & Learning**: Teach how agents make decisions, plan under uncertainty, and adapt through learning. Decision processes: utility theory, Markov Decision Processes (MDPs); Planning under uncertainty; Reinforcement learning fundamentals (MDP, policy, reward shaping); Deep RL essentials (PPO, DQN, A3C — as applicable); Meta-cognition and adaptation in agents; Ethical and value-aligned decision making
UNIT-III (10 Hrs)	**Large Language Models & Prompt-Driven Agent Behavior:** Cover how modern agentic systems leverage large language models and advanced prompt techniques. Transformer architecture and attention basics; Role of LLMs in agentic systems; Prompt engineering fundamentals (zero-shot, chain-of-thought, ReAct); Retrieval-Augmented Generation (RAG) and memory systems; Context management and embedding techniques; Tool-usage and action invocation via prompts
UNIT-IV (10 Hrs)	**Tools, Frameworks & System Design:** Introduce hands-on frameworks used to build agentic AI and how to compose multiple agents into workflows. Overview of agent frameworks: LangChain, Lang Graph, AutoGen, CrewAI (documentation & practice); Memory, embedding databases (e.g., FAISS, Pinecone, ChromaDB); Workflow orchestration and multi-agent coordination; Agent-to-agent communication; Deployment patterns and APIs; Monitoring, scalability, and production concerns
UNIT-V (10 Hrs)	**Responsible & Applied Agentic AI:** Address real-world applications, ethical considerations, safety, and career perspectives. Responsible agentic AI design and risk mitigation; Bias, fairness, and interpretability in autonomous systems; Privacy, transparency, and accountability. Safety and alignment challenges; Case studies: Healthcare, cybersecurity, business automation

Textbooks:	
1.	Stuart J. Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, 4th Ed., Pearson,2022.
2.	Michael Wooldridge, An Introduction to MultiAgent Systems (2nd ed.), 2011
3.	Richard S. Sutton &Andrew G. Barto — Reinforcement Learning: An Introduction, 2nd Edition, Jain Book Depot, 2018
References	
1.	David L. Poole & Alan K. Mackworth — Artificial Intelligence: Foundations of Computational Agents, 3rd Edition, Cambridge University Press, 2023
2.	Ken Huang (Ed.) — Agentic AI: Theories and Practices (Springer, 2025)

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AM4109	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

EDGE AI

(For AIML)

Course Objectives:

1.	To learn the techniques and components of Edge Artificial Intelligence.
2.	To apply AI knowledge to develop Edge Artificial Intelligent Systems.
3.	To find optimized solutions for given problems.

Course Outcomes: At the end of the course, Students will be able to

S.No	Outcome	Knowledge Level
1.	Apply edge computing concepts and architectures to industrial applications and challenges**.**	**K3**
2.	Apply edge computing, virtualization, DevOps, and CI/CD tools to build and manage basic deployment pipelines.	**K3**
3.	Apply segmentation techniques to improve efficiency of AI models and develop secured distributed Edge applications.	**K3**
4.	Apply knowledge of AI for optimizing Edge application	**K3**
5.	Understand and apply concepts of Mobile Edge AI	**K3**

SYLLABUS

UNIT-I (10Hrs)	**Fundamentals of Edge Computing**: Introduction, Need, Key Techniques, Benefits, Systems Paradigms of Edge computing, Edge Computing Frameworks, Value Scenarios for Edge Computing. Edge computing system architectures. Industrial Applications of Edge Computing, Intelligent Edge and Edge Intelligence, Challenges and opportunities in Edge Computing.
UNIT-II (10 Hrs)	**Role in Edge Computing:** A high-level hardware hierarchy of edge computing paradigm, Virtualization: Virtual Machine and Container, Network Virtualization, Introduction to DevOps: Understanding the history and evolution, Overview of the benefits and challenges, Introduction to DevOps tools and practices, **CI/CD**: tools such as Jenkins, Travis CI, and CircleCI and CircleCI, GitLab CI/CD Setup and Configuration, installing and setting up GitLab CI/CD, Configuring GitLab CI/CD pipelines, Creating a basic CI/CD configuration file (.gitlab-ci.yml)
UNIT-III	**Artificial Intelligence Inference in Edge:** Optimizing AI models in Edge: General

(10 Hrs)	method, Edge device, Overview of TensorFlow Lite (TFLite) format and its benefits, Introduction to Open Neural Network Exchange (ONNX) format and its advantages, Understanding NVIDIA TensorRT format and its optimizations for inference Segmentation of AI Model, Segmentation of AI Model, Early Exit of Inference (EEoI), Sharing of AI Computation**.**
UNIT-IV (10 Hrs)	**Artificial Intelligence for Optimizing Edge:** AI for Adaptive Edge Caching: use cases DNNs and DRL, Optimizing Edge Task Offloading, Edge Management and Maintenance: Communication, security, joint Edge optimization.
UNIT-V (10 Hrs)	**Mobile Edge AI** Overview, Edge inference: On-device inference, Computation offloading, Server-based edge inference, Device-edge joint inference, Edge training: Data partition-based, Model partitionbased, Coded computingTechnology aspects, Agiling capabilities, Tool stack implementation, People aspect, processes

Textbooks:

1.	Wang, X., Han, Y., Leung, V. C., Niyato, D., Yan, X., & Chen, X.," Edge AI: Convergence of edge computing and artificial intelligence", Singapore: Springer,2020, ISBN 978-981-15-6185-6
2.	Jie Cao, Quan Zhang, Weisong Shi "Edge Computing: A Primer", Springer International Publishing
3.	Mobile Edge Artificial Intelligence Opportunities and Challenges By Yuanming Shi, Kai Yang, Zhanpeng Yang, Yong Zhou 2021, ISBN - 9780128238172, 0128238178, Elsevier Science publication

Reference Books:

1.	Pethuru Raj et al., *Applied Edge AI: Concepts, Platforms, and Industry Use Cases*, CRC Press.
2.	Parikshit N. Mahalle et al., *Edge Artificial Intelligence: Algorithms, Applications, Challenges and Ethical Issues*, Elsevier.
3.	Javid Taheri et al., *Edge Intelligence: From Theory to Practice*, Springer.
4.	Junaid Shuja et al., *AI at Edge: Transforming Edge Networks with Computational Intelligence*, Springer.

e-Resources

1.	Liu, Deyin; Chen, Xu; Zhou, Zhi; Ling, Qing (2020). HierTrain: Fast Hierarchical Edge AI Learning With Hybrid Parallelism in Mobile-Edge- Cloud Computing. IEEE Open Journal of the Communications Society

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AM4111	**SEC**	**--**	**1**	**2**	**2**	**30**	**70**	**3 Hrs.**

PROMPT ENGINEERING

(For AIML)

Course Objectives:

1	Provide students with practical knowledge of prompt engineering techniques for effective interaction with Large Language Models.
2	Enable students to design, refine, and apply prompt-based methods for generating structured and reasoning-oriented outputs.
3	Develop students’ ability to build retrieval-augmented and prompt-driven AI applications using modern tools and frameworks.
4	Equip students to evaluate LLM-based and multimodal applications with respect to accuracy, safety, ethics, and robustness.

Course Outcomes: After successful completion of the course, the student will be able to:

S. No.	Outcome	Knowledge Level
1	Apply iterative and advanced prompting techniques to obtain accurate, context-aware, and task-specific responses from LLMs	K3
2	Use prompts for producing structured outputs and reasoning-based responses for different tasks	K3
3	Build retrieval-augmented generation pipelines using LangChain workflows, embeddings, and vector stores	K3
4	Evaluate prompt-driven agents and multimodal applications with respect to response quality, safety, ethics, and robustness.	K5

SYLLABUS

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Unit I: Foundations of Prompt Engineering: Definition of prompt engineering, Distinction between prompt engineering and model fine-tuning, Motivation and benefits of prompt engineering, Core principles of effective prompt design**,** Anatomy of a prompt, Setting up the Python environment for LLM interaction, Iterative prompting lifecycle, Common prompt pitfalls and remediation

Sample Experiments:

1. Environment & Connectivity: Install required packages (e.g., transformers, openai); securely configure the API key; run a simple "Hello, world" prompt to verify model access.
2. Baseline vs. Enhanced Prompts: Execute a naïve prompt ("Write a one-paragraph bio of Ada Lovelace.") and an enhanced prompt that adds role framing, specificity, and explicit format instructions; compare both outputs for relevance, completeness, and style.
3. Iterative Refinement on a Simple Task: Summarize the plot of the Shakespearean play Romeo and Juliet in two sentences through three rounds of prompt tweaking:
   a. Minimal instruction.
   b. Addition of length and style constraints
   c. Specification of key content elements (setting and theme)

   Document how each iteration changes and improves the result.
4. Diagnosing Prompt Failures & Edge Cases: Craft a vague or contradictory prompt; analyze the failure mode (ambiguity, missing context, or format errors); refine the prompt by adding examples or clarifying instructions

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2	**Unit II: Advanced Prompt Patterns & Techniques:** Enhanced prompt anatomy: contextual detail and explicit output specifications, Few-shot in-context prompting, Prompt structuring and template design, Role-based prompting to establish personas or system behavior, Negative prompting to filter or suppress undesired content, Constraint specification and instruction enforcement (e.g., length, format), Iterative prompt refinement and optimization **Sample Experiments:** 1. Few-Shot vs. Zero-Shot Comparison: Design and execute a zero-shot prompt and a few- shot prompt (with 2–3 exemplar input-output pairs) for a chosen text task (e.g., sentiment classification or translation); compare outputs for accuracy, consistency, and adherence to examples. 2. Role-Based & Negative Prompting: Craft a role-based prompt to establish a specific persona (e.g., “You are a financial advisor…”); then create a negative prompt to suppress undesired content (e.g., “Do not mention any brand names”); evaluate how each influences the model’s response. 3. Constraint Specification & Iterative Refinement: Select an open-ended task (e.g., summarizing a technical article); issue a basic prompt; identify failures in length or format; refine the prompt by adding explicit constraints (word count, bullet format, etc.); document improvements over two refinement cycles.
3	**Unit III: Structured Output & Reasoning Techniques:** Importance of structured outputs for real-world applications, Prompting for specific formats (lists, tables, Markdown), Generating valid JSON and YAML via explicit instructions, Eliciting chain-of-thought reasoning in zero-shot prompts, Decomposing complex tasks into manageable sub-tasks **Sample Experiments:** 1. Structured Format Prompting: Instruct the model to output information as bullet lists and Markdown tables (e.g., “List three benefits of daily exercise in a Markdown table with columns ‘Benefit’ and ‘Description.’”); verify the output matches the requested structure. 2. JSON/YAML Generation: Provide a brief dataset description (e.g., three books with title, author, publication year) and prompt the model to produce valid JSON or YAML; use a parser to validate syntax and refine the prompt if errors occur. 3. Chain-of-Thought & Task Decomposition: Present a multi-step problem (e.g., a logic puzzle) and apply zero-shot CoT prompting (e.g., “Let’s think step by step. Explain your reasoning before the final answer.”); separately, decompose the problem into sequential sub-questions, collect partial answers, combine them, and compare accuracy against a direct-answer baseline

4	**Unit IV: Retrieval-Augmented Generation &LangChain Workflows:** Limitations of LLM internal knowledge, Need for external data sources, Introduction to Retrieval-Augmented Generation (RAG), Overview of RAG architecture (indexing vs. retrieval + generation), Getting started with LangChain for LLM applications, Basics of LangChain Expression Language (LCEL), Simplified indexing pipeline: document loading & text splitting, Fundamentals of embeddings and vector stores, Building a basic retrieval-generation pipeline with an LCEL chain **Sample Experiments:** 1. Building a Simple LCEL Chain: Create a minimal LCEL script that accepts a fixed instruction (e.g., “Summarize this text: …”), passes it to an LLM, and prints the result; verify end-to-end execution. 2. Basic Data Indexing for RAG: Load a small collection of documents; split into uniform chunks (e.g., 200 tokens); generate embeddings for each chunk; store them in an in- memory vector store; inspect for consistency. 3. Constructing & Running a Basic RAG Chain: Build a pipeline that: a. Receives a user query b. Retrieves the top-k relevant chunks c. Constructs a combined prompt with context + query d. Send it to the LLM e. Returns the answer Test with sample queries and compare factual accuracy against a prompt without retrieval.

5	**Unit V: Agents, Multimodal AI & Ethical Evaluation:** Introduction to LLM agents and their basic architecture, Overview of multimodal AI models (VLMs), Prompting for text-to-image generation and image understanding, Importance of prompt evaluation beyond subjective judgment, Manual evaluation techniques (heuristic checks for accuracy, relevance, format), Introduction to “LLM-as-Judge” for automated evaluation, Security considerations (prompt injection, sensitive-information risks), Prompt-based mitigation strategies for safety and robustness, Ethical concerns (bias, misinformation, data privacy), Brief exploration of UI frameworks (Streamlit/Gradio) for deploying prompt-driven apps, Adapting to the evolving nature of prompt engineering through continuous learning **Sample Experiments:** 1. Building a Simple LLM Agent: Register a tool (e.g., a calculator function) and craft prompts that instruct the agent to invoke it when required; implement using LangChain or a function-calling API; test on queries requiring tool execution. 2. Multimodal Prompting Exploration: Generate images from detailed text prompts; feed one generated image into an image-understanding model or API with an appropriate prompt; compare the returned caption to the original prompt to evaluate alignment. 3. Prompt Evaluation & Ethics Workshop: a. Select two existing prompts and generate multiple outputs; apply manual heuristic checks for accuracy, relevance, and format compliance. b. Use an “LLM-as-Judge” prompt (e.g., “Rate these outputs on a scale of 1–5 for clarity and correctness.”) to automate evaluation. c. Design a prompt- injection test (e.g., “Ignore previous instructions…”), observe the response, then refine system prompts to mitigate the vulnerability.
Reference Books:	
1	Phoenix, James, and Mike Taylor. Prompt engineering for generative AI. " O'Reilly Media, Inc.", 2024.
2	Nathan Hunter, The Art of Prompt Engineering with ChatGPT: GPT-4, Plugins & DALL·E 3 Update, October 2023.
3	AI Prompt Engineering: Foundations of Communication with LLMs – Building Generative AI and Agentic AI Prompt Systems Across Development, Testing, and Deployment (AI Engineering, Nelson Ming, Ashford Partners, 2025
e-Resources	
1	Prompt Engineering Guide, https://www.promptingguide.ai
2	https://developers.openai.com/api/docs/guides/prompting
3	https://github.com/openai/openai-cookbook
4	https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for-developers/
6	“Prompt Engineering for Generative AI”, Google, https://cloud.google.com/discover/what-is-prompt-engineering

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC4101	MC	2	--	--	--	30	--	--

CONSTITUTION OF INDIA

(Common to all)

Course Objectives:

1.	To provide understanding of the fundamental principles, sources, and key features of the Indian Constitution.
2.	To familiarize students with the structure and functioning of Union, State, and Local Governments in India.
3.	To develop awareness about the roles of constitutional bodies like the Election Commission and welfare commissions in promoting democracy and social justice.

Course Outcomes: At the end of the course, students will be able to

S.No.	Outcome	Knowledge Level
1.	Apply the concepts of the Indian Constitution to interpret its features such as Citizenship, Preamble, Fundamental Rights and Duties, and Directive Principles of State Policy.	K3
2.	Apply the structure and functions of the Union Government to analyze the roles of executive, legislature, and judiciary in the Indian administrative system.	K3
3.	Apply the structure and functions of the State Government to interpret the roles of the Governor, Chief Minister, and State Secretariat in administration.	K3
4.	Apply the structure and functions of local administration to interpret the roles of district, municipal, and Panchayati Raj institutions in grassroots governance.	K3
5.	Apply the roles and functions of the Election Commission and various welfare commissions to interpret their significance in ensuring democratic governance and social justice.	K3

SYLLABUS

UNIT-I (6 Hrs)	Introduction to Indian Constitution: Constitution meaning of the term, Indian Constitution - Sources and constitutional history, Features - Citizenship, Preamble, Fundamental Rights and Duties, Directive Principles of State Policy.

UNIT-II (6 Hrs)	Union Government and its Administration Structure of the Indian Union: Federalism, Centre- State relationship, President: Role, power and position, PM and Council of ministers, Cabinet and Central Secretariat, Lok Sabha, Rajya Sabha, The Supreme Court and High Court: Powers and Functions.
UNIT-III (6 Hrs)	State Government and its Administration Governor - Role and Position - CM and Council of ministers, State Secretariat: Organisation, Structure and Functions.
UNIT-IV (6 Hrs)	Local Administration - District's Administration Head - Role and Importance, MunicipalitiesMayor and role of Elected Representative - CEO of Municipal Corporation Pachayati Raj: Functions PRI: Zilla Panchayat, Elected officials and their roles, CEO Zilla Panchayat: Block level Organizational Hierarchy - (Different departments), Village level - Role of Elected and Appointed officials - Importance of grass root democracy.
UNIT-V (6 Hrs)	Election Commission: Election Commission- Role of Chief Election Commissioner and Election Commissionerate State Election Commission: Functions of Commissions for the welfare of SC/ST/OBC and women.

Textbooks:	
1.	Durga Das Basu, Introduction to the Constitution of India, Prentice — Hall of India Pvt. Ltd..New Delhi.
2.	Subash Kashyap, Indian Constitution, National Book Trust.
Reference Books:	
1.	J.A. Siwach, Dynamics of Indian Government & Politics, Sterling Publishers.
2.	D.C. Gupta, Indian Government and Politics, S Chand.
3.	H.M.Sreevai, Constitutional Law of India, 4th edition in 3 volumes, Universal Law Publication.
4.	M.V. Pylee, Indian Constitution Durga Das Basu, Human Rights in Constitutional Law, Prentice Hall of India Pvt. Ltd.. New Delhi.
5.	Noorani, A.G., (South Asia Human Rights Documentation Centre), Challenges to Civil Right), Challenges to Civil Rights Guarantees in India, Oxford University Press 2012.

e-Resources	
1.	https://onlinecourses.swayam2.ac.in/e-learning/preview/imb26_mg151
2.	https://onlinecourses.nptel.ac.in/noc24_lw05/preview
3.	https://nptel.ac.in/courses/129106750

Regulation: R23	IV / IV - B.Tech. II - Semester								
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course code	**Course Name**	**Category**	**L**	**T**	**P**	**Cr**	**C.I.E**	**S.E.E**	**Total Marks**
B23AM4201	Project and Internship	**PR**	**-**	**-**	**24**	**12**	**60**	**140**	**200**

Course Code	Category	L	T	P	C	C.I.E	S.E.E	Exam
B23AM4201	PR	--	--	24	12	60	140	3H

PROJECT WORK

For AIML

Pre-requisites: SE and Technical Skills in different Technologies, skills, planning, documentation

Course Objectives: The main objectives of the course are to

1.	To provide an opportunity to work in group on a topic / problem / experimentation
2.	To encourage creative thinking process
3.	To provide an opportunity to analyze and discuss the results to draw conclusions
	To acquire and apply fundamental principles of planning and carrying out the work plan of the project through observations, discussions and decision-making process

Course Outcomes: At the end of the course, students will be able to

S.No.	Outcome	Knowledge Level
1.	Identify a current problem through literature/field/case studies	K3
2.	Identify the objectives and methodology for solving the problem	K3
3.	Design and Develop technology/process for solving the problem	K4
4.	Evaluate the technology/process	K5

SYLLABUS

*The object of Project Work is to enable the student to take up investigative study in the broad field of Artificial Intelligence and Machine Learning , either fully theoretical/practical or involving both theoretical and practical work to be assigned by the Department on an individual basis or a group of students, under the guidance of a Supervisor.

This is expected to provide a good initiation for the student(s) in R&D work.

The assignment to normally include:

a) Survey and study of published literature on the assigned topic.

b) Working out a preliminary approach to the problem relating to the assigned topic.

c) Conducting preliminary Analysis/Modelling/Simulation/Experiment/Design/ Feasibility.

d) Preparing a written report on the study conducted for presentation to the department.

e) Final Seminar, as oral Presentation before a departmental committee